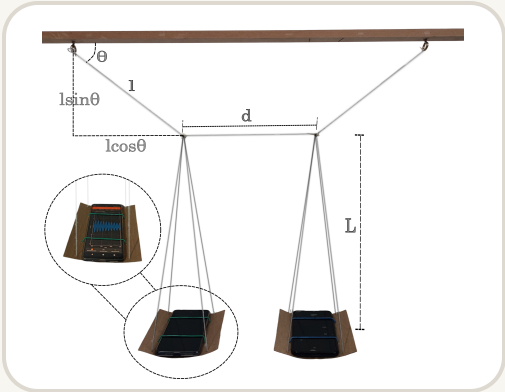


Experimental Study of Two Coupled Pendulums using FFT and Mobile Sensors

Hugo Sebastián Ramírez Cutimbo

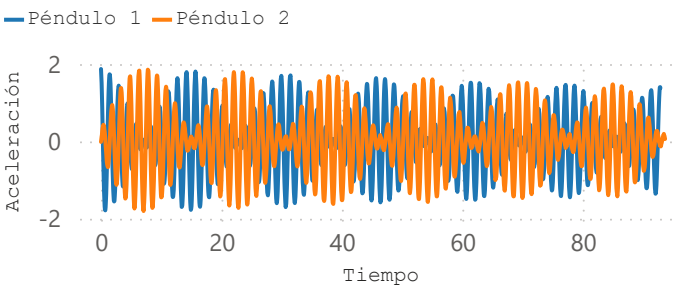
THEORETICAL FRAMEWORK

- The behavior of the coupled pendulum system shown in the figure was experimentally studied.
- Acceleration data were acquired using the mobile application phyphox.
- The Fast Fourier Transform (FFT) of the recorded signals was obtained using the NumPy library in Python.

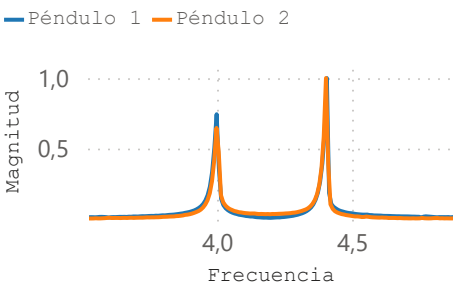


EXPERIMENTAL RESULTS

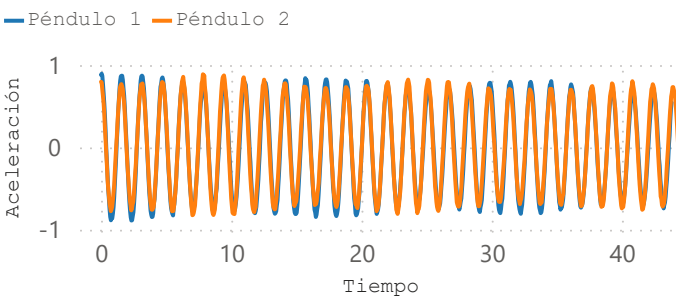
Beating Mode



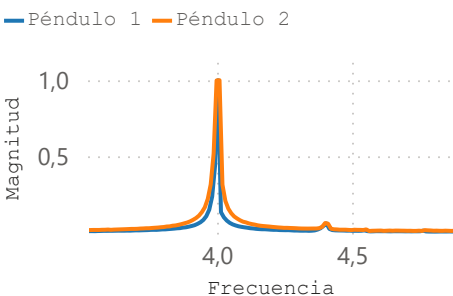
Beating Mode FFT



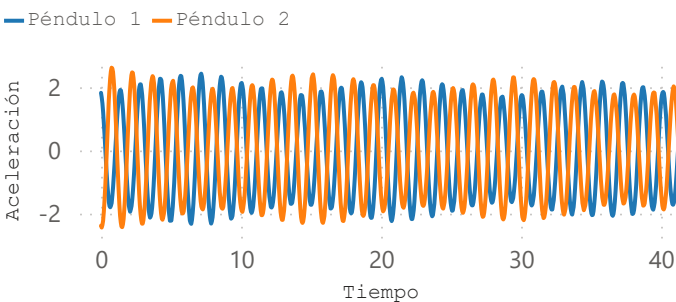
Symmetric Mode



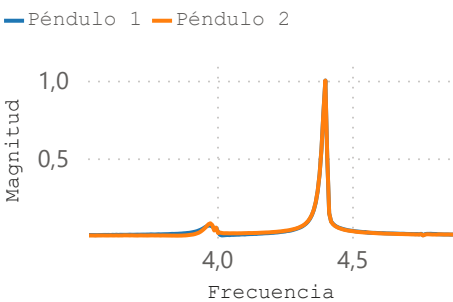
Symmetric Mode FFT



Antisymmetric Mode



Antisymmetric Mode FFT



MODE	Freq. 1 [rad/s]	Freq. 2 [rad/s]
Antisymmetric		4,40
Beating	4,00	4,40
Symmetric	4,00	

CONCLUSIONS

- For each studied case, the motion can be decomposed into a linear combination of the symmetric and antisymmetric modes.
- Each motion, regardless of its initial conditions, can be reconstructed as a function of the frequencies obtained through the Fast Fourier Transform (FFT).